Arithmetic, Answer Formatting, and Generalization in a Tiny Language Model Trained with GRPO

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Abstract

This project investigates direct reinforcement learning on a pretrained 0.8-billion-parameter language model for Countdown arithmetic. A pilot run improved strict greedy accuracy from 0/256 to 85/256 (33.2%) on a fixed held-out set after 300 optimizer steps. This report distinguishes measured findings from planned experiments. A post-hoc audit using a conservative declared-expression parser recovers the same correct counts when the answer-tag requirement is relaxed. This supports investigating arithmetic task improvement, while not establishing general reasoning or fully identifying the role of formatting. Output-budget, multi-seed, reward-ablation, and distribution-shift experiments are tracked below as they are completed.

1 Question and experimental scope

When does direct GRPO improve arithmetic task performance, and how much of an apparent improvement is answer-format compliance? The base checkpoint is `Qwen/Qwen3.5-0.8B-Base`; the dataset is `Jiayi-Pan/Countdown-Tasks-3to4`. The experiment omits supervised fine-tuning. It does not train a model from scratch and does not claim discovery of general reasoning. Model and dataset revisions, code, and outputs are retained in the public repository.

2 Pilot protocol

The text backbone is trained with BF16 rank-16 LoRA on one RTX 4090 Laptop GPU with 16 GB VRAM. GRPO uses eight candidates per puzzle, 32 completions per optimizer step, a learning rate of 10^{-5}, no KL penalty, and a 256-token completion limit. The 300-step run used seed 42 and took 5,135.9 seconds of training time (85.6 minutes), excluding setup and evaluation.

Puzzles are deduplicated by $(\text{target}, \text{sort}(\text{numbers}))$. A seeded hash partition reserves 2% for evaluation before subset selection. The available training subset contains 20,000 puzzles; evaluation uses 256 fixed puzzles. Both base and adapter evaluations use greedy decoding with the same output limit. No evaluation-based checkpoint selection was performed.

The strict reward is binary: one final `<answer>` block must contain a valid arithmetic expression using each supplied number exactly once. The verifier supports binary $+, -, \times, /$, parentheses, and exact rational arithmetic. It rejects number reuse/omission and unsafe syntax. Negative and fractional intermediate values are allowed. No formatting bonus was used.

3 Pilot results

Metric on 256 held-out puzzles	Base	GRPO adapter
Strict correct	0 (0.0%)	85 (33.2%)
Strict valid expressions	11 (4.3%)	90 (35.2%)
Declared-expression diagnostic correct	0 (0.0%)	85 (33.2%)

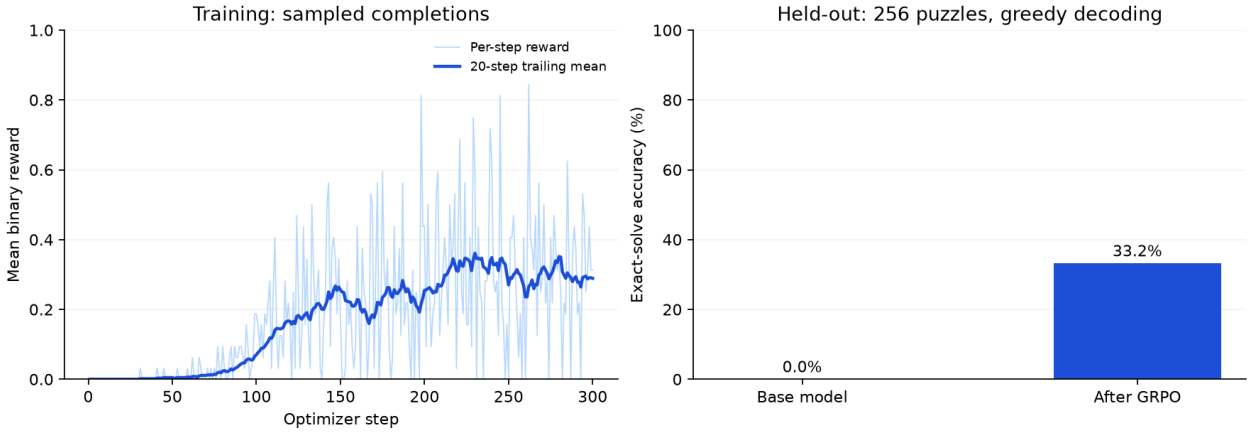


Figure 1: Pilot stochastic training reward and separate greedy held-out accuracy. These use different puzzles and decoding protocols; validation was not measured at every training step.

The first positive training reward occurred at step 31. Peak PyTorch-allocated memory was 2.72 GiB, which excludes reserved memory and other GPU allocations. In the final 30 training steps, 59.9% of sampled completions reached the output cap. Historical evaluation reports do not record termination token IDs, so their truncation status is unknown rather than inferred from decoded text.

4 Format-independent diagnostic: declared-expression-v1

This diagnostic is post-hoc and is not used as a training reward. Extraction is deterministic and independent of the target:

1. If answer tags occur, require exactly one complete block and extract it. A wrong tagged answer never falls back to an earlier expression.
2. Otherwise, select the last standalone arithmetic line, allowing an optional "Answer:" or "Final answer:" prefix and simple math/code delimiters.
3. For an equation, evaluate only the leftmost expression. Do not trust its claimed right-hand value or search intermediate calculations for a success.
4. Apply bounded AST interpretation, exact rational evaluation, and multiset validation of input numbers. Reject unsupported operations and syntax.

This conservative parser can miss correct answers expressed in prose or other notation. Therefore equal strict and diagnostic counts do not prove formatting had no causal role in training. They show that this specified relaxation did not recover additional correct answers from the archived generations.

Diagnostic outcome	Base	Adapter
Correct	0	85
Wrong numerical value with valid numbers	90	5
Number multiset mismatch	152	121
Invalid syntax	10	19
No declared expression	4	24
Ambiguous or incomplete answer tags	0	2

These are mutually exclusive diagnostic categories. Number omission/reuse details are retained per answer; truncation is a separate flag when token metadata exists. All 512 historical outputs were re-scored, and strict scores matched their original records. The same unique holdout puzzles were verified in both reports, with no intersection with the selected training subset. An independent exact subset-dynamic-programming solver found a valid witness for all 256 puzzles under the all-inputs-once rules. Unsolvable holdout items therefore do not explain the errors. Witnesses are archived separately and never included in model prompts. An agent review of the first two records per observed outcome category (20 records total) found agreement with the specified extraction rule; this is not a blinded human annotation study. Prose-heavy and incomplete outputs illustrate the parser's coverage limitations.

5 Controlled study design and current status

The study keeps the original strict metric alongside the diagnostic. Planned comparisons are not presented as completed results:

- Matched base/adapter output budgets of 256, 512, and 1024 tokens; record actual generated lengths, EOS status, truncation, and generation time.
- Binary correctness versus a 0.05 bonus for valid incorrect expressions, with training seeds 42, 43, and 44 and a fixed split seed of 42.
- Three-number-only training with disjoint three- and four-number evaluation.
- Separately labeled, deterministic, solver-verified suites with larger inputs and out-of-range targets; solutions are never included in prompts.
- A comparison demo with exact verifier feedback and archived examples.

Per-seed results and held-out uncertainty will be reported separately. Four distribution-shift/arity suites have been frozen before querying trained models.

Uncertainty and paired analysis. Individual checkpoint accuracies use 95% Wilson score intervals, following the NIST formulation. For the pilot's 85/256 score this interval is 27.7–39.2%. It describes puzzle-sampling uncertainty under a binomial model, not training-seed variability or uncertainty about arbitrary out-of-distribution tasks. Comparisons match canonical puzzles and report gained, lost, jointly correct, and jointly incorrect answers. Exact two-sided McNemar tests are exploratory, with unadjusted p-values; we do not use the grid to claim a confirmatory significance result.

Partial output-budget result. The trained adapter solved 86/256 puzzles at 512 tokens, compared with 85/256 at 256 tokens: one gained solution and no lost solutions. At 512 tokens, 140/256 outputs (54.7%) still hit the cap. Strict and diagnostic scores coincide. The 512-token evaluation therefore provides little evidence that doubling the output budget resolves the pilot's failures. The 1024-token cells are still pending; this is not the final budget comparison.

6 Provenance and experiment ledger

The pilot used clean source commit `0a432a272591f7683b711a9c867c01e7d939fca1`. W&B contains source snapshots, configs, package versions, puzzle splits, checkpoints, and generation tables:

- Pilot baseline
- Pilot training
- Pilot final evaluation
- Final pilot adapter artifact
- Study workspace
- Archived-output diagnostic audit

2026-10-10. Inspected the current repository, pilot artifacts, local GPU, and Overleaf access. Recorded the approved research plan. Implemented and tested the declared-expression diagnostic, re-scored archived outputs, and added token termination metadata and output-budget overrides to evaluation. Twenty-five CPU tests initially passed; solver coverage increased this to 29 tests. Created and compiled this Overleaf report, verified all holdout puzzles are solvable, and reviewed a deterministic diagnostic sample. The matched output-budget grid is running. Six controlled training runs are queued on a Colab L4; the first is live. Its initial five steps averaged approximately 50.5 seconds per step using the reference PyTorch attention implementation. These are preliminary throughput measurements, not a completed training runtime. Statistical analysis and pairing validation bring CPU coverage to 36 passing tests.

7 Limitations

One seed and 256 puzzles do not establish robust generalization. Training-dataset deduplication cannot rule out exposure in pretraining. Strict success conflates arithmetic correctness with protocol compliance; the diagnostic is an imperfect post-hoc measure, not a causal decomposition. Long answers may contain useful work but still fail due to truncation. No cross-seed, longer-output, or OOD claims are made until the corresponding experiments finish.